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# Opportunities, challenges, and future directions for simulation modeling the effects of structural racism on cancer mortality in the United States: a scoping review

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## Abstract

**Purpose:** Structural racism could contribute to racial and ethnic disparities in cancer mortality via its broad effects on housing, economic opportunities, and health care. However, there has been limited focus on incorporating structural racism into simulation models designed to identify practice and policy strategies to support health equity. We reviewed studies evaluating structural racism and cancer mortality disparities to highlight opportunities, challenges, and future directions to capture this broad concept in simulation modeling research.

**Methods:** We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses-Scoping Review Extension guidelines. Articles published between 2018 and 2023 were searched including terms related to race, ethnicity, cancer-specific and all-cause mortality, and structural racism. We included studies evaluating the effects of structural racism on racial and ethnic disparities in cancer mortality in the United States.

**Results:** A total of 8345 articles were identified, and 183 articles were included. Studies used different measures, data sources, and methods. For example, in 20 studies, racial residential segregation, one component of structural racism, was measured by indices of dissimilarity, concentration at the extremes, redlining, or isolation. Data sources included cancer registries, claims, or institutional data linked to area-level metrics from the US census or historical mortgage data. Segregation was associated with worse survival. Nine studies were location specific, and the segregation measures were developed for Black, Hispanic, and White residents.

**Conclusions:** A range of measures and data sources are available to capture the effects of structural racism. We provide a set of recommendations for best practices for modelers to consider when incorporating the effects of structural racism into simulation models.

Simulation modeling is a powerful computational tool that could be used to inform the development of cancer care policies, guidelines, clinical trials, and web-based personalized clinical decision tools (1-5). In this context, the simulation models developed by the National Cancer Institute Cancer Intervention and Surveillance Modeling Network have provided data to inform cancer screening policies and guidelines in the United States (1,2,6-8).

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National Cancer Institute Cancer Intervention and Surveillance Modeling Network models comparing model results across racial groups acknowledge that race is a sociopolitical construct that reflects the impact of racism on cancer outcomes. Currently, the models capture the effects of racism via its race-specific input parameters on cancer incidence, screening and treatment dissemination, stage distributions, survival, and competing mortality (9–11). The models assume that the underlying effects of structural racism are reflected in the racial differences in cancer burden (eg, incidence) and care delivery (eg, screening and treatment). Although individual and structural racism are conceptualized as the root cause of racial disparities (12), the models have not yet incorporated measures of structural racism or its direct impact on cancer processes and outcomes (13). Simulation modeling structural racism could potentially provide useful data to inform policies and practices that could eliminate structural racism and promote equity in cancer care. However, prerequisite to such efforts, the models would require information on measurement and real-world data linking structural racism to disease processes captured in the models, including its effects on cancer mortality.

The definition of structural racism has evolved over the last decade (14–19). A review published by Bailey et al. (18) provides a contemporary definition for investigating the impact of structural racism on cancer outcomes (20). Structural racism is "the totality of ways in which societies foster racial discrimination through mutually reinforcing systems of housing, education, employment, earnings, benefits, credit, media, health care, and criminal justice" (18). Although, structural racism has been reviewed in the literature (17,21), there have been no systematic or scoping reviews on structural racism in the context of simulation modeling research. We addressed this gap by conducting a scoping review of peer-reviewed articles evaluating the impact of structural racism on cancer mortality–related racial and ethnic disparities in the United States. The overall findings were used to provide recommendations for best practices to incorporate the effects of structural racism into simulation models. This study is intended to highlight the opportunities, challenges, and future directions for the development of novel simulation models that will inform equitable cancer care policies and guidelines in the United States.

## Methods

Our scoping review was conducted following a standard methodological framework (22–24). A scoping review was considered for this study as our goal was to scope a body of literature, clarify concepts and measures, and provide recommendations for best practices to incorporate the effects of structural racism into simulation modeling research (25). The standard methodological framework (22–24) provides 6 stages to guide scoping review processes including specifying the research question; identifying relevant literature; selecting studies; data mapping; summarizing, synthesizing, and reporting the results; and including expert consultation. The review was carried out in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (26) (Supplementary Table 1, available online), and the study was registered in Open Science Framework (doi: 10.17605/OSF.IO/SP8TH) (27). Since we included a review of published articles and study-level results, this study did not require submission for institutional review board approval or exemption.

The following definitions of race, ethnicity, structural racism, and cancer mortality were used to develop the search strategy for this scoping review. Race is a social construct rooted in the historical, social, political, and economic contexts (28). Ethnicity refers to the social and cultural characteristics, backgrounds, or experiences shared by a group of people (29). We identified self-reported Black or African American, American Indian or Alaska Native, Asian, Native Hawaiian or other Pacific Islander, White, Middle Eastern or North African, and Hispanic or Latino/a racial and ethnic groups according to the Office of Management and Budget standards (30). Structural racism is defined above (18). Cancer mortality included death due to cancer and/or all-cause mortality in individuals diagnosed with cancer (31).

Guided by these definitions, we searched for studies evaluating the impact of structural racism and its dimensions (ie, housing, education, employment, earnings [income], benefits, credit, health care, media, and criminal justice) on racial and ethnic disparities in cancer mortality. A search strategy was developed using an iterative approach, with 3 rounds of preliminary searches, and refinement of the search strategy based on initial search results by a trained librarian (GB) at the National Institutes of Health. Each preliminary search was performed in the PubMed–MEDLINE database constrained to English-only language restrictions and research published between January 1, 2018, and January 1, 2023, to inform current policies considering modern cancer care. The comprehensive search strategy included a combination of keywords, synonyms, medical subject headings, and Emtree terms related to structural racism, race, ethnicity, and cancer mortality. Three researchers were consulted to gauge the suitability of the search terms (Supplementary Table 2, available online). The searches were performed across the following 6 widely used databases: MEDLINE via PubMed (National Library of Medicine), PsycINFO (American Psychological Association), Embase (Elsevier), Scopus (Elsevier), Web of Science Core Collection (Clarivate Analytics), and Cochrane CENTRAL (Wiley & Sons). Additional references were extracted from reviews, systematic reviews, and meta-analyses. Duplicates were omitted.

### Eligibility criteria

Eligibility criteria were based on preliminary searches and included primary empirical research studies published between January 2018 and January 2023; involving human participants, samples, and/or datasets; specific to the US population or setting; in any type of cancer (except skin cancer) and cancer mortality; and with the terms *race and ethnicity* and *structural racism and/or its dimensions* in the title or abstract. Further details are provided in Supplementary Table 3 (available online).

### Data screening, extraction, and synthesis

The review was conducted in Covidence, a web-based publication screening software platform. Titles and abstracts of the articles were screened independently by 3 researchers (JJ, KW, JW). Any discrepancies in the inclusion or exclusion decision were resolved through discussion. A team of extractors independently performed full-text screening of articles.

Data extraction was conducted independently using an online form, which was pilot tested and customized to support the review by ensuring charting consistency and inter-reviewer reliability. The data extraction template included the title, purpose, research design, target population and setting, inclusion and exclusion criteria, start and end date, site and type(s) of cancer, data sources, primary outcome, race and ethnicity categories,

individual characteristics, variables used to capture structural racism and/or its dimensions, statistical methods, results, limitations of the study, funding source(s), and conflicts of interest. We also extracted the definitions and conceptual frameworks used to guide the research question(s) and variable selection in the studies. Data were summarized descriptively in tabular form and text using a narrative approach.

## Results

A total of 8345 records were identified through the search. After removing duplicate or ineligible articles, 1038 articles underwent full text review, and 183 articles were included in the final review (Figure 1). Most studies were designed to evaluate mortality disparities across White (26%), Black (25%), Hispanic (19%), or Asian or Pacific Islander (18%) groups, whereas relatively fewer studies focused on American Indian or Alaska Native groups (3%) and we found no studies on Middle Eastern or North African groups (Supplementary Table 4, available online).

### Definitions, theories, and conceptual frameworks

A total of 40 studies mentioned theories and conceptual frameworks to guide the research question(s) and variable selection (Table 1). Intersectionality Theory describing the interconnected nature of social identities and/or positions was used to explore the interactions between race and ethnicity and structural factors (n = 2) (32–36). Eco-social Theory of Disease Distribution (37) (n = 3) (38–40) and Urban Theory (41) (n = 1) (42) were used for variable selection. For example, the variables for Concentrated Disadvantage Index (42) was selected based on Urban Theory (41) regarding the overconcentration of Black individuals, children, and female-headed families in poor neighborhoods.

In our sample of studies, structural racism was commonly (n = 20) measured using racial residential segregation (39,40,43–60). Racial residential segregation is defined as the physical separation of the racial groups in residential contexts (61). Segregation could lead to potentially worse socioeconomic and physical conditions for minoritized groups at the individual, household, neighborhood, and community levels (45,62,63).

### Measures

The 183 studies evaluating the impact of structural racism on cancer mortality used 150 measures of structural racism, including socioeconomic (housing, education, employment, and income; n = 113), health care and benefits (n = 34); combined socioeconomic and health care (n = 2); and criminal justice (n = 1) dimensions. Accordingly, 114 measures were related to social and economic policy, 34 measures were related to health-care policy, and 2 measures were related to a combination of social, economic, and health policies (12). The measures, single or composite, descriptions, structural racism dimensions, individual or

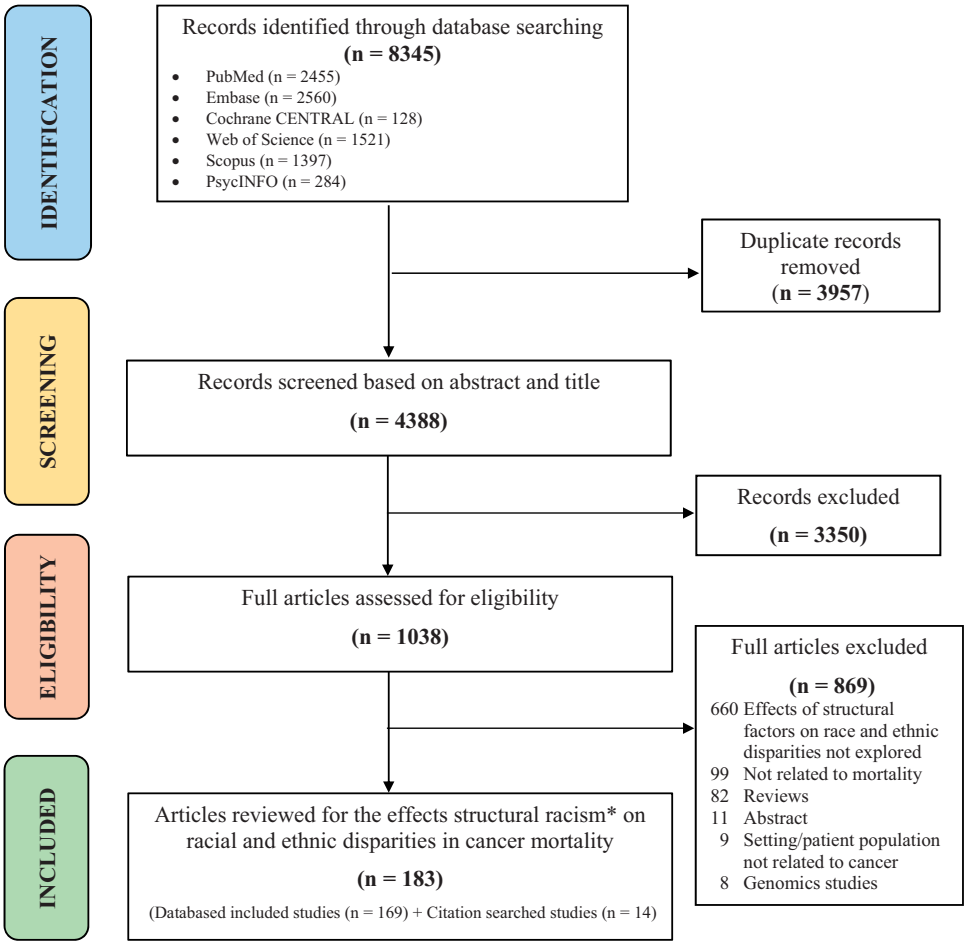


**Figure 1.** Article identification process.

*Structural racism* was defined as the totality of ways in which societies foster racial discrimination through mutually reinforcing systems of housing, education, employment, earnings, benefits, credit, media, health care, and criminal justice (18).

**Table 1.** Summary of definitions, theories, and conceptual frameworks mentioned in the studies included in the scoping review

| Concept | Description |
|---|---|
| Structural racism | "The ways in which mutually reinforcing systems of housing, education, employment, earnings, benefits, credit, media, health care, and criminal justices are root causes of health disparities" (18).<br>"Structural racism refers to a system in which public policies, institutional practices, cultural representations, and other norms work to perpetuate racial group inequity" (83). |
| Racial residential segregation | Segregation refers to the physical separation of the races in residential contexts (61). Williams et al. highlights that "segregation was imposed by legislation, supported by major economic institutions, enshrined in the housing policies of the federal government, enforced by the judicial system, and legitimized by the ideology of white supremacy that was advocated by churches and other cultural institutions. These institutional policies combined with the efforts of vigilant neighborhood organizations, discrimination on the part of real estate agents and home sellers, and restrictive covenants to limit the housing options of Black Americans to the least desirable residential areas" (61,84–86). Racially segregated areas in the United States are primarily minoritized neighborhoods with persistent poverty, low home equity, and stagnant social mobility (46,87). |
| Housing discrimination | Housing discrimination limits high-quality housing availability to individuals of a given race (18,88). Housing discrimination can take several forms, including biased mortgage lending, discrimination in renting, and racial segregation (55). |
| Intersectionality | Intersectionality theory posits that dimensions of structural inequality, such as racism and classism, are mutually constitutive (ie, intersectional), therefore, social groups at the nexus of more than 1 marginalized identity experience unique forms of disadvantage (34–36). |
| Eco-social Theory of Disease Distribution | Health disparities in various disease outcomes are a product of social, ecological, political, and historical exposures across multiple levels, including neighborhoods (37,44,89,90). |
| Urban Theory | The Urban Theory observes the economic, social, and physical barriers of urban settings seen in economically disadvantaged communities (41). |
| Berkman framework on social integration and health | This theoretical framework serves to identify social networks contributing to breast cancer survival. Social networks are explored in continuum, with each stage acting as a precursor to the next—macro social context, network structure, and mobilized network resources, respectively (91,92). |
| A framework proposed by Kirtane and Lee (93) | Racial disparities are driven by biological factors, individual factors, health behaviors, and structural barriers (93,94). |
| National Institute on Minority Health and Health Disparities Framework | Conceptualizes various levels of influence at the individual, interpersonal, community, and societal levels with biological, behavioral, sociocultural and environmental, and health-care system domains of influence over the course of one's lifetime to better understand the factors that influence health disparities in minority populations (95). |
| Social inequalities along the childhood cancer continuum | A framework model developed to study the impact of equity in childhood cancer by connecting structural determinants of health inequality in childhood cancer outcomes with social determinants of childhood cancer (96). |

geographic unit of analysis, and data sources are provided in Supplementary Table 5 (available online). Composite measures were created using structural equation modeling or principal component analysis.

There was variation in the measures used to capture structural racism. For example, racial residential segregation was measured using the Dissimilarity Index (n = 11); Index of Concentration at the Extremes (n = 6); Redlining Index (n = 2); Isolation Index (n = 2); historical, georeferenced Homeowners Loan Corporation grading data (n = 1); and a composite measure of structural racism incorporating tract-level racial segregation, disadvantage, and affluence (n = 1) (Supplementary Table 5, available online) (39,40,43–60). Index of Dissimilarity measured segregation by assessing the heterogeneity of 2 racial groups within a geographical area (45–49,51,53,57). The Index of Concentration at the Extremes is a measure of spatial social polarization quantifying extremes of deprived and privileged social groups, and it consists of 3 measures encompassing economic segregation (income-based Index of Concentration at the Extremes), racial segregation (race-based Index of Concentration at the Extremes), and racialized economic segregation (race income–based Index of Concentration at the Extremes) (64). Overall, these measures of segregation and structural racism consisted of area-level variables belonging to the housing, income, education, employment, and/or credit dimensions (39,40,43–60). Health-care (eg, health care insurance, facilities) dimensions were not directly reflected in the measures of racial residential segregation.

The health-care (and benefits) dimension(s) included individual-level (eg, insurance status, travel distance), facility-level (eg, facility type, accreditation), and area-level (eg, county-level screening rate, state Medicaid expansion) measures. At least 2 studies considered composite measures combining socioeconomic and health-care dimensions (eg, California Healthy Places Index) (65,66). In our sample of studies, there were no composite measures capturing all the dimensions of structural racism in a single variable. Further, we did not find measures of structural racism specific to American Indian or Alaska Native, Native Hawaiian, or other Pacific Islander, or Middle Eastern or North African groups (eg, isolation scores for Middle Eastern or North African).

### *Data sources and unit of analysis*

Data sources included cancer registry, hospital, or survey data commonly linked to US census data for area-level characteristics. Data were collected at the individual, facility, census tract, block, city or town, and county levels.

### *Analytic approaches*

In the 183 studies, we identified 6 distinct methods used to quantify the effects of structural racism (its dimensions) on racial and ethnic disparities in cancer mortality: 1) racial and ethnic disparities in a single location (eg, rural hospital, urban city); 2)

comparison of subgroups defined by race and ethnicity and a structural factor (eg, rural Asian vs urban White individuals); 3) effects of sequential adjustment of structural factors (with other demographic and clinical variables) on racial and ethnic disparities in multivariable models; 4) structural factors as potential mediators of racial and ethnic disparities in mortality; 5) interactions between structural factors and race and ethnicity and effect modification; and 6) geospatial analyses to identify structural factors associated with area-level race or ethnicity-specific mortality rates.

### *Impact of structural racism on racial and ethnic disparities in cancer mortality*

The results varied by cancer site, data sources, methods, geographic unit of analysis, location, racial and ethnic comparators, structural racism measure(s), and other covariates included in the analysis (Supplementary Table 6, available online). We illustrate the variation in results using residential segregation as an example (Table 2) (39,40,43–60).

High residential segregation measured by the Index of Dissimilarity was associated with worse survival in Black residents with pancreatic, breast, colorectal, oral, lung, liver, and prostate cancer (n = 7) (45–49,51,57) but not among White residents with breast, colorectal, or oral cancer (n = 3) (48,49,51). Homeowners Loan Corporation grading data in New Jersey showed that residence in areas graded as "best" was associated with lower breast cancer–specific mortality among non-Latina White women (hazard ratio [HR] = 0.48, 95% confidence interval [CI] = 0.35 to 0.65) (59). In fully adjusted race-stratified models, Hispanic women with epithelial ovarian cancer showed increased hazard of overall mortality in neighborhoods with extreme economic segregation (HR = 1.41, 95% CI = 1.18 to 1.69) and racialized economic segregation (for non-Hispanic Black HR = 1.42, 95% CI = 1.10 to 1.82; for Hispanic HR = 1.34, 95% CI = 1.03 to 1.74) (44). Another study found segregation measured by Index of Concentration at the Extremes was associated with worse breast cancer survival in White and Black but not in Hispanic residents (60). Adjustment for the tract-level composite measure of structural racism (in addition to age, sex, and diagnosing hospital) reduced the cancer mortality disparity for Hispanic patients from a hazard ratio of 1.20 (95% CI = 0.83 to 1.72) to 0.96 (95% CI = 0.65 to 1.43) and from 1.61 (95% CI = 1.15to 2.26) to 1.04 (95% CI = 0.65 to 1.66) for non-Hispanic Black patients with acute myeloid leukemia (58). Approximately half of the studies (n = 9) were location specific, and the segregation measures were developed for Black, Hispanic, and White residents.

## Discussion

Simulation modeling is a methodological tool used to inform cancer care policies and guidelines; therefore, incorporating the impact of structural racism into simulation modeling research could potentially help advance equity in cancer care. The 183 studies included in this review showed wide variation in the measures, data sources, methods, and results. We found 150 measures belonging to 8 dimensions of structural racism including housing, education, employment, income, credit, benefits, health care, and criminal justice. The segregation measures captured area-level socioeconomic and credit dimensions of structural racism.

A previous systematic review conducted by Groos et al. (17) in 2018, summarizing 20 articles evaluating the impact of structural racism on the health and well-being of individuals also noted the heterogeneity in the measurement of structural racism. This variation may arise from the complexity as well as the multidimensional and multilevel impact of structural racism on individual health (17,67). Recent studies have provided practical guidance and recommendations for measuring and incorporating the effects of structural racism in health-care research (20,67,68). These recommendations may also be useful for simulation modeling research. For example, a recent report by the Assistant Secretary for Planning and Evaluation (68) provided a broad overview of existing measures and methods for considering area-level structural inequalities. Further, studies also recommend the use of index measures, developing measures targeted to capture structural racism in other racial and ethnic groups, and use of psychometric evaluations to test measures for relevance over historical eras and life course (20).

### Strengths and limitations of the scoping review

Our study was developed and reviewed by a multidisciplinary research team. The search strategy included 6 electronic bibliographic databases to ensure a broad search of the literature. The inclusion and exclusion criteria, as well as the data extraction forms were pilot tested and revised prior to the implementation of the study. However, our results should be reviewed within the context of the limitations of this study. This review did not include studies outside of the United States and articles published in languages other than English. Therefore, our findings may not reflect global patterns of the role of structural racism (or its equivalent) in other countries. We also did not include studies published prior to January 2018 as our goal was to inform model development considering modern cancer care. Moreover, the systematic review by Groos et al. (17) included studies published up to 2017, albeit the goals were different to the current study. We limited the scope of this study to review the effects of structural racism on cancer mortality disparities. Although this approach helped us identify the next steps needed for modeling, the downstream impact of structural racism on cancer mortality could also be studied based on its impact on cancer natural history, screening and timely diagnosis, and treatment separately.

### Simulation modeling the effects of structural racism

Simulation models are developed using input parameters that capture the life course of individuals from birth to death including the natural history of cancer and risk factor exposures (eg, smoking) (6). Screening performance and dissemination parameters are overlaid on the life course and natural history for early detection of precancerous lesions or preclinical cancers, whereas treatment efficacy and dissemination parameters are used to alter postdiagnosis survival rates. The effects of structural racism could be incorporated into these models via its impact on natural history, screening dissemination, and treatment efficacy and/or dissemination. The data gaps report by Trentham-Dietz et al. (13) highlights the need for data on the effects of structural racism, adequate sample size for precise parameter estimation, and multidisciplinary collaboration for content knowledge and lived experiences of those affected by structural racism. In our study, we identified several opportunities, challenges, and future directions to support the incorporation of structural racism into population models.

### Opportunities

The input parameters for structural racism could be developed using publicly available data sources outlined in Supplementary Table 5 (available online) (eg, US Census). Ideally, such parameters should be developed in collaboration with the scientists

**Table 2.** Summary of studies evaluating the impact of racial residential segregation (one component of structural racism) on racial and ethnic disparities in cancer mortality

| Manuscript title | Disease site | Data source | Variables used to measure racial segregation | Statistical methods | Primary findings |
|---|---|---|---|---|---|
| Residential racial segregation and disparities in breast cancer presentation, treatment, and survival (48) | Breast | SEER and US Census American Community Survey | County-level Index of Dissimilarity[a] | Multivariable Cox proportional hazards model | In adjusted models:<br>• Black patients had a 29% increased hazard of overall stage–specific mortality (HR = 1.29, 95% CI = 1.04 to 1.60) with increasing Index of Dissimilarity.<br>• White patients had no statistically significant relationship with Index of Dissimilarity and death (HR = 1.13, 95% CI = 0.97 to 1.33). |
| Structural racism and breast cancer-specific survival: impact of economic and racial residential segregation (60) | Breast | Cancer registries across 2 hospital systems:<br>NCI-designated cancer center and a partner safety-net hospital in South Florida | Census tract–level income and race-based Index of Concentration at the Extremes[b] | Multilevel Cox models | In the fully adjusted models stratified by race and ethnicity:<br>• Non-Hispanic White residents in low-income and racially segregated neighborhoods showed increased hazard of breast cancer–specific death compared with non-Hispanic White residents living in high-income neighborhoods ($HR_{Income}$ = 2.40, 95% CI = 1.40 to 4.13; $HR_{NHB}$= 2.13, 95% CI = 1.16 to 3.92; $HR_{Hispanic}$ = 2.24, 95% CI = 1.35 to 2.70; $HR_{IncomeNHB}$ = 4.61, 95% CI = 2.69 to 7.85; $HR_{IncomeHispanic}$ = 3.08, 95% CI = 1.85 to 5.09).<br>• Non-Hispanic Black residents in low-income neighborhoods had an increased hazard of death (HR = 2.22, 95% CI = 1.10 to 4.50).<br>• Hispanic residents showed no statistically significant association between any form of segregation and breast cancer mortality. |
| Racialized economic segregation and breast cancer mortality among women in Maryland (52) | Breast | Maryland Cancer Registry and US Census | Census tract–level Index of Concentration at the Extremes[b] (income, race, and race and income-based Index of Concentration at the Extremes) | Age- and race-adjusted Cox models (accounting for clustering at the census tract–level) | Results varied by the measure of Index of Concentration at the Extremes, race (Black and White), age (younger than 60 years; 60 years and older), prognostic factors, and time since diagnosis. |
| The impact of neighborhood economic and racial inequalities on the spatial variation of breast cancer survival in New Jersey (43) | Breast | New Jersey State Cancer Registry and US Census | Census tract–level Index of Concentration at the Extremes[b] (income, race, and race and income-based Index of Concentration at the Extremes) | Nonlinear geo additive Bayesian models | The percentage of geographic disparity based on spatial variance decreased from 89.7% in the null model to 24.7% in the model including age, stage, subtype, race and ethnicity, marital status, and insurance type and further reduced to 15.3% when income based Index of Concentration at the Extremes was included in the models. |
| Mortgage lending bias and breast cancer survival among older women in the United States (54) | Breast | SEER–Medicare and Home Mortgage Disclosure Act data | Redlining Index[c] identifies regions within the same metropolitan statistical areas (within SEER) containing properties that are less likely | Multivariable Cox proportional hazards model | • No linear relationship between Redlining Index and mortality.<br>• The adjusted relationship between race and all-cause and breast cancer mortality with Redlining Index was not statistically significant. |

(continued)

**Table 2.** (continued)

| Manuscript title | Disease site | Data source | Variables used to measure racial segregation | Statistical methods | Primary findings |
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| | | | to secure mortgages. | | • Women living in more redlined areas experienced poorer all-cause and breast cancer–specific survival, with the most pronounced effect among women without comorbid conditions. |
| Association between residence in historically redlined districts indicative of structural racism and racial and ethnic disparities in breast cancer outcomes (59) | Breast | New Jersey State Cancer Registry and historical, georeferenced Homeowners Loan Corporation grading data | HOLC grades: best, hazardous, still desirable, definitely declining | Multivariable cox proportional hazards model | • The associations between HOLC grade varied by race and ethnicity.<br>• Compared with residence in HOLC redlined areas, residence in HOLC areas graded "best" was associated with lower hazard of breast cancer–specific death (HR = 0.48, 95% CI = 0.35 to 0.65) but only among non-Latina White women. |
| Structural racism is a mediator of disparities in acute myeloid leukemia outcomes (58) | Blood | Patient data from 6 academic cancer centers in the metropolitan Chicago region and American Community Survey | Structural racism[d] at the census tract–level | Mediation analysis with discrete-time models | Adjustment for census tract–level structural racism (in addition to age, sex, and hospital) reduced the disparities for leukemia death in:<br>• Hispanic patients (vs non-Hispanic White patients) from a hazard ratio of 1.20 (95% CI = 0.83 to 1.72) to 0.96 (95% CI = 0.65 to 1.43).<br>• Non-Hispanic Black patients (vs non-Hispanic White patients) from a hazard ratio of 1.61 (95% CI = 1.15 to 2.26) to 1.04 (95% CI = 0.65 to 1.66). |
| The impact of racial residential segregation on colorectal cancer outcomes and treatment (49) | Colorectal | SEER and US Census American Community Survey | County-level Index of Dissimilarity[a] | Multivariable Cox proportional hazards model | • Black patients experienced a 43% increase in 5-year cancer-specific mortality with increasing Index of Dissimilarity (HR = 1.43, 95% CI = 1.17 to 1.74).<br>• White patients had no statistically significant relationship with Index of Dissimilarity (HR = 0.99, 95% CI = 0.87 to 1.11). |
| Racial residential segregation and colorectal cancer mortality in the Mississippi Delta region (50) | Colorectal | SEER and US Census Bureau | Racial residential segregation for White and Black residents in each Delta county using the multilevel index of dissimilarity (97) | Mixed linear regression models | County-level colorectal cancer mortality rates among Black and White residents in Delta region counties varied by rurality and White–Black residential segregation measured by multilevel index of dissimilarity. |
| Ecological study of variability in the relationship between liver cancer mortality and racial residential segregation (56) | Liver | Wisconsin Department of Health Statistics' Vital Records Service and US Census | Census tract–level Racial Residential Segregation[e] measured using local Black and Hispanic isolation scores (98) | Geospatial analysis of 5 metropolitan statistical areas in Wisconsin | • Higher Black isolation was associated with increased likelihood of a tract being in a high-mortality cluster and decreased likelihood of being in a low-mortality cluster (driven by 2 of 5 metropolitan statistical areas).<br>• No clear exposure–response relationship for Hispanic isolation. |
| The role of racial segregation in treatment and outcomes among patients with hepatocellular carcinoma (47) | Liver | SEER and US Census American Community Survey | Index of Dissimilarity[a] calculated within 100 most populous participating SEER counties | Kaplan–Meier method and competing risk models | 5-year cancer-specific (overall and by stage) and overall survival by Index of Dissimilarity for Black and White patients. |

(continued)

**Table 2.** (continued)

| Manuscript title | Disease site | Data source | Variables used to measure racial segregation | Statistical methods | Primary findings |
|---|---|---|---|---|---|
| The impact of residential racial segregation on non-small cell lung cancer treatment and outcomes (57) | Lung | SEER and US Census American Community Survey | County-level racial Index of Dissimilarity[a] | Competing risk models | In the adjusted models:<br>• Black patients in the highest quartile of Index of Dissimilarity had 5% increased hazards (ie, 5-year cancer-specific death) compared with White patients in the highest quartile of Index of Dissimilarity (HR = 1.05, 95% CI = 1.03 to 1.08) and 8% increased hazards compared with White patients in the lowest quartile of Index of Dissimilarity (HR = 1.08, 95% CI = 1.05 to 1.12;).<br>• Black patients in the lowest quartile of Index of Dissimilarity had 4% increased hazards of death compared with White patients in the lowest quartile of Index of Dissimilarity (HR = 1.04, SE = 1.02, 95% CI = 1.01 to 1.08).<br>• White patients in the highest quartile of Index of Dissimilarity had 3% increased hazards of death compared with White patients in the lowest quartile of Index of Dissimilarity (HR = 1.03, 95% CI = 1.01 to 1.05). |
| Effects of racial residential segregation on oral squamous cell carcinoma prognosis and survival (51) | Oral | SEER and US Census American Community Survey | County-level Index of Dissimilarity[a] | Multivariate Cox proportional hazards model | • Extreme Index of Dissimilarity score was associated with increased overall death in Black individuals (HR = 1.42, 95% CI = 1.19 to 1.70).<br>• No association with Index of Dissimilarity in White individuals (HR = 1.01, 95% CI = 0.96 to 1.07). |
| Residential segregation and overall survival of women with epithelial ovarian cancer (44) | Ovarian | Florida Cancer Data System and US Census American Community Survey | Census-defined place-based Index of Concentration at the Extremes[b]:<br>• Economic segregation (income-based Index of Concentration at the Extremes)<br>• Racial segregation (race-based Index of Concentration at the Extremes for non-Hispanic Black and Hispanic populations)<br>Racialized economic segregation (race income–based Index of Concentration at the Extremes) | Multilevel Cox models | Living in economically marginalized areas was associated with statistically significantly increased (all-cause) death after controlling for individual-level factors:<br>• Economic segregation for marginalized vs privileged neighborhoods (HR = 1.15, 95% CI = 1.06 to 1.25).<br>• Racial segregation for more concentrated non-Hispanic Black segregation (HR = 1.12, 95% CI = 1.02 to 1.22), and more Hispanic segregation (HR = 1.02, 95% CI = 0.93-1.11) vs concentrated non-Hispanic White neighborhoods.<br>• Racialized economic segregation for non-Hispanic Black low-income neighborhoods (HR = 1.21, 95% CI = 1.12 to 1.32) and low-income Hispanic segregated |

(continued)

**Table 2.** (continued)

| Manuscript title | Disease site | Data source | Variables used to measure racial segregation | Statistical methods | Primary findings |
|---|---|---|---|---|---|
| | | | | | neighborhoods (HR = 1.12, 95% CI = 1.03 to 1.22) vs non-Hispanic White high-income neighborhoods.<br>• Fully adjusted race-stratified models showed that Hispanic women had an increased hazard of death in all forms of segregation. |
| The impact of residential segregation on pancreatic cancer diagnosis, treatment, and mortality (53) | Pancreas | SEER and US Census | County-level racial Index of Dissimilarity[a] was calculated from the census block–level estimates of the Black and White population within the county. | Kaplan–Meier method | Kaplan–Meier estimates of overall survival and stage-specific survival (up to 12 mo) stratified by race for the lowest and highest quartiles of segregation. |
| Socioeconomic mediation of racial segregation in pancreatic cancer treatment and outcome disparities (45) | Pancreas | SEER and US Census | County-level Index of Dissimilarity[a] | Mediation analysis using generalized structural equation modeling | Black patients in the highest levels of Index of Dissimilarity reported an 8% increase in overall death (HR = 1.08, 95% CI = 1.03 to 1.14) compared with White patients in the lowest level of Index of Dissimilarity. |
| The impact of racial residential segregation on prostate cancer diagnosis and treatment (46) | Prostate | SEER and US Census American Community Survey | Index of Dissimilarity[a] calculated within 100 most populous participating SEER counties | Cumulative hazards and Fine–Gray competing risk models | 10-year overall and cancer-specific cumulative hazards for overall and by stage for Black and White patients in the highest and the lowest quartiles of Index of Dissimilarity. |
| Trends in mortality among Black and White men with prostate cancer in Massachusetts and Pennsylvania: race and neighborhood socioeconomic position (40) | Prostate | Massachusetts and Pennsylvania cancer registries and US Census | Neighborhood socioeconomic position, a summary score generated using census tract–level measures of % poverty, median family income, % households with female head, % with income from interest or rent, % White, % Black, % unemployment, % male unemployment, % receiving welfare assistance, and the joint race and income-based Index of Concentration at the Extremes[b] | Multivariable Cox proportional hazards model | Disparities associated with neighborhood socioeconomic position were either stagnant or widened across all endpoints (ie, all-cause, prostate cancer, and cardiovascular mortality) among men with prostate cancer from 2000 to 2015. |
| Using the index of concentration at the extremes at multiple geographical levels to monitor health inequities in an era of growing spatial social polarization: Massachusetts, USA (2010-14) (39) | Cancer (all-site) | Massachusetts Vital Statistics | Census tract and city or town level Index of Concentration at the Extremes[b] (income, race, and race and income-based Index of Concentration at the Extremes) | Multilevel Poisson models for age-standardized mortality rates | Cancer (all-site) mortality rate ratios for Index of Concentration at the Extremes measures (income, race, and race and income-based) overall and by census tract, city, and town level, gender, and race and ethnicity. |

(continued)

**Table 2.** (continued)

| Manuscript title | Disease site | Data source | Variables used to measure racial segregation | Statistical methods | Primary findings |
|---|---|---|---|---|---|
| Housing discrimination and racial cancer disparities among the 100 largest US metropolitan areas (55) | Cancer (all-site) | CDC WONDER and NCHS and US Census and SEER and Home Mortgage Disclosure Act data | Redlining Index[c] (Metropolitan Statistical Area); Dissimilarity Index (Black) (census tract); Isolation Index (Black)[f] (census tract) | Geographic patterns and correlations | Correlations between structural racism measures and Black-to-White cancer mortality (rate ratios) disparities by gender. |

[a] A validated measure of segregation. It assesses the heterogeneity of 2 racial groups within a geographical area with values ranging from zero (complete integration) to 1 (complete segregation) and represents the proportion of people within that area who would need to move to achieve complete integration. CDC WONDER = Centers for Disease Control and Prevention Wide-ranging ONline Data for Epidemiologic Research; CI = confidence interval; HOLC = Homeowners Loan Corporation; HR = hazard ratio; NCHS = National Center for Health Statistics; NCI = National Cancer Institute; SEER = Surveillance Epidemiology and End Results; NHB = non-Hispanic Black.

[b] A measure that quantifies the extent to which persons in a specified area are concentrated into the top vs bottom extremes of a specified social distribution. The Index of Concentration at the Extremes was initially a measure of economic residential segregation and was expanded to racial or ethnic and racialized economic segregation to capture the extremes of high-income White communities and low-income Black or Hispanic communities. As a result, Index of Concentration at the Extremes captures both racial or ethnic and economic elements of residential segregation at the same time. Economic segregation was defined as the difference in the number of persons living in households earning no more than $25 000 compared with the number of persons living in households earning $100 000 or more as a proportion of the total population with household income information; racial segregation was defined as the difference in the number of Black or Hispanic persons compared with the number of White persons as a proportion of the total persons with race and ethnicity information; and racialized economic segregation was defined as difference in the number of Black or Hispanic persons in households earning no more than $25 000 compared with White persons in households earning $100 000 or more as a proportion of the total persons with income and race or ethnicity information.

[c] Measures the odds ratio of denial of a mortgage application based on the property location, comparing local properties with all properties within the same metropolitan statistical areas. This index identifies regions within metropolitan statistical areas containing properties that are less likely to secure mortgages. The index was estimated using logistic regression modeling in an adaptive spatial filtering framework based on Home Mortgage Disclosure Act data.

[d] A composite variable incorporating racial segregation, disadvantage, and affluence. Racial segregation measured by the proportion of individuals who were non-Hispanic Black and proportion of individuals who were non-Hispanic White; disadvantage was measured by the proportion of families with incomes below the poverty line, proportion of families receiving public assistance, proportion of adults who were unemployed, and proportion of households that were female-headed households with children; and *affluence* was defined by the proportion of families with incomes higher than $75 000, proportion of adults with at least a college education, and proportion of adults employed in professional or managerial occupations.

[e] Measures how much racial and ethnic demographics in a sub area (census-tract) deviate from the expected case that all racial and ethnic groups are equally distributed across a region.

[f] The percentage of same-group population in the geographic location where the average member of a racial and ethnic group lives.

previously involved in creating the original measure. The PhenX Toolkit (69) Social Determinants of Health Project (70) provides a collection of publicly available measurement protocols to facilitate collaboration and combination of study data to increase statistical power.

Structural equation modeling techniques could potentially be used to develop novel composite measures that capture the multiple dimensions related to structural racism (71,72). These measures may provide additional insight into the mechanisms underlying population health inequities. Further, US Census data have been available for several decades, which allows modelers to capture the variation of structural factors over time in specific geographic locations. Moreover, several advanced methods have been developed to geocode patient locations and link individual data to broader area-level characteristics (73). There are a growing number of variables for income, housing, education, employment, credit, benefits, and health-care characteristics that fall within the definition of or are influenced by structural racism. This, combined with novel statistical methods to develop new composite measures, provides novel opportunities to explore the incorporation of structural racism into simulation modeling research.

## Challenges

The variation of measures and methods evaluating the effects of structural racism poses a challenge in summarizing results across studies for input parameter development. The use of area-level characteristics (eg, zip code–level median income) derived from US Census data as proxies for individual-level characteristics (eg, income) could lead to ecological fallacy (74,75). Ecological fallacy refers to the incorrect assumption that the associations observed at the aggregated level are necessarily the same at the individual level (76). Studies have shown that the effects from aggregate data could be incorrect in magnitude and direction at the individual level (74–76). Also as noted in the data gaps report (13), other challenges include limited generalizability of data to inform national-level policies, standardization of data collected over long periods of time, data missingness, and lack of structural measures specific to other minoritized populations (eg, American Indian, Alaska Native), which could be necessary to capture modern trends in cancer incidence and mortality in the United States (77,78).

## Future directions for modeling

To our knowledge, currently, there are no simulation models that directly model the effects of structural racism on cancer outcomes. Current models consider race as a proxy for the effects of racism. Therefore, based on our findings, we provided recommendations for best practices to incorporate the effects of structural racism into cancer simulation models (Table 3). Our recommendations were developed considering the components provided in the International Society for Pharmacoeconomics and Outcomes Research and the Society for Medical Decision Making (ISPOR-SMDM) Modeling Good Practices Task Force recommendations (79).

The effects of structural racism in minoritized communities have been studied for decades by researchers belonging to different fields and disciplines (eg, historians). Therefore, it is important to collaborate with multidisciplinary health equity researchers and partner with minoritized communities when developing new definitions, guidelines, and measures to model the effects of structural racism. Further, modeling should consider the impact of using individual-level (eg, income) vs area-level variables (eg, neighborhood median income) of structural factors. Survey data collected at the individual level (when available) could be less prone to ecological fallacy (75).

**Table 3.** Recommendations for best practices to incorporate the effects of structural racism into simulation models

| Simulation model components | Recommendations for best practices |
|---|---|
| Model conceptualization | • Consult early and often with health equity experts (and other stakeholders) to ensure that the goals and model development represent the impact of structural racism on cancer processes.<br>• Evaluate the incorporation of expert and stakeholder feedback.<br>• Develop an explicit process a priori to incorporate the impact of structural racism into model structure ensuring that the model reflects current knowledge.<br>• The conceptualization of the effects of structural racism should not be driven by data availability.<br>• Develop clear written statements of the decision problem, objectives, and scope to ensure accessibility to the multidisciplinary stakeholders. |
| Measures of structural racism | • Develop measures for structural racism in consultation with health equity experts and other stakeholders considering the decision problem, cancer type, target population, historical era, and context.<br>• Evaluate and consider the variation of the impact of the selected measures of structural racism by race and ethnicity (eg, the effects of racial residential segregation).<br>• Carefully consider the variation of the meaning and interpretation of a given measure (eg, income) based on the use of single (eg, income) vs composite measures (eg, a socioeconomic status index) and individual-level (eg, individual income) vs area-level (eg, neighborhood median income) variables.<br>  • The use of single vs composite measures should be guided by the decision problem and context. Composite measures could capture the multiple dimensions of structural racism, and single measure may provide actionable policy levers.<br>  • Evaluate the impact of ecological fallacy (75) when using area-level measures.<br>  • Input parameter development for area-level effects should account for clustering using appropriate statistical methods (eg, multilevel modeling, latent variable analysis).<br>• Consider the effects of the structural measure(s) along the life course of individuals and its varying effects at different historical time points in collaboration with health equity scholars.<br>• Ensure that the meaning and definitions of variables remain consistent when multiple data sources are merged to develop input parameters that could track the effects of structural measures over time.<br>• Evaluate observational data, meta-analysis, and other sources of data used for input parameter development for quality and methodological rigor. |
| Incorporation of the effects of structural racism into simulation modeling | • Consider modeling the effects of structural racism on physiological systems (eg, allostatic load, inflammation, oxidative stress, DNA damage) that contribute to cancer when data are available.<br>• Use age-period cohort models stratified by the structural measure(s) considering the role of history and birth cohort effects of structural racism.<br>• Include the effects of structural racism on the likelihood (probability) of receiving screening. These effects could be incorporated via screening dissemination models stratified by the structural factor(s).<br>• Include the effects of structural factors on diagnostic follow-up (eg, time to biopsy) and treatment delays. These could be incorporated using the effects of structural racism on the likelihood (probability) of receiving treatment via treatment dissemination models stratified by the structural factor(s).<br>• Use real-world treatment effectiveness for addressing social, economic, and health-care policy–relevant issues. Treatment efficacy could be adjusted using a decrement that quantifies the effects of structural racism. The decrements could be derived using meta-analyses or observational data. |
| Uncertainty | • Clearly explain the sources of uncertainty related to the measure of structural racism.<br>• Describe the impact of uncertainty in the measure of structural racism on model output.<br>• Provide results for a range of values (eg, confidence interval) pertaining to the measure(s) of structural racism.<br>• Evaluate the impact of using different data sources on the effects of structural racism. |
| Transparency and validation | • Provide documentation on the input parameter and model development in culturally relevant and appropriate lay language.<br>• Make input parameters, methods, and programs accessible to all stakeholders.<br>• Clearly outline the assumptions associated with the incorporation of structural racism.<br>• Use an independent data source representative of the target population for model validation. |

Modeling should consider the effects of structural racism along the life course of individuals and across different historical eras.

Currently, there is no single data source that could provide all the relevant individual, clinical, and structural measures needed to comprehensively study disparities in cancer mortality. Modeling could combine information from different data sources and provide a virtual laboratory to better understand the effects of structural racism and other factors on racial and ethnic disparities in cancer mortality. Recent studies have shown that targeted interventions to increase screening among minoritized populations; timely resolution of abnormal findings; initiation and completion of therapy; and patient navigation could potentially help eliminate disparities in colorectal cancer mortality (80–82). In this context, simulation modeling could be used as a tool to study the downstream effects of structural racism along the life course and design hypothetical targeted interventions to eliminate disparities in cancer outcomes. Our study provides recommendations for best practices to consider when developing these simulation models.

## Data availability

All the studies summarized in this scoping review are available in the article and in its online supplementary material.

## Author contributions

Jinani Jayasekera, PhD, MS (Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Project administration; Resources; Supervision; Validation; Writing—original draft; Writing—review & editing), Chiranjeev Dash, MBBS, PhD (Resources), Marquita Lewis-Thames, MPH, MS, PHD (Funding acquisition; Resources), Shakira Grant, MBBS (Funding acquisition; Resources), Christina Chapman, MD (Writing—review & editing), Gisela Bethea, MEd, MLIS (Methodology; Validation; Writing—review & editing), Amrita Ramanathan, BA (Data curation; Formal analysis; Writing—original draft; Writing—review & editing), Randy Le, BS (Data curation; Formal analysis; Methodology), Julia Zhang, BA (Data curation; Formal analysis; Methodology), Dalya Kamil, BSc (Data curation; Formal analysis; Writing—original draft; Writing—review & editing), Traci Bethea, MPA, PhD (Funding acquisition; Resources), Stephanie Ponce, BA (Data curation; Formal analysis; Writing—original draft; Writing—review & editing), Che-Jung Chang, PhD, MS (Data curation; Writing—original draft; Writing—review & editing), Kemi Ogunsina, MD, PhD (Data curation; Writing—original draft; Writing—review & editing), Manami Bhattacharya, PhD, MS (Data curation; Writing—original draft; Writing—review & editing), Jennifer Ish, PhD, MS (Data curation; Writing—original draft; Writing—review & editing), Adaora Ezeani, MD, MPH (Data curation; Writing—review & editing), Jennifer Woo, PhD, MPH (Data curation; Formal analysis; Writing—original draft), Safa El Kefi, PhD (Writing—review & editing), Kaitlyn Wojcik, MPH (Conceptualization; Data curation; Formal analysis; Writing—original draft), Jessica Fernandez, PhD (Writing—original draft; Writing—review & editing), Camryn Cohen, BS (Data curation; Formal analysis; Writing—original draft; Writing—review & editing), and Allana Forde, PhD, MPH (Writing—review & editing).

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## Conflicts of Interest

None.

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